Modular Arithmetic and the Euclidean Algorithm

Mingxun Zhou (mingxunz@ust.hk)

COMP2711 Discrete Mathematical Tools

1 Integer division and divisibility

Modular arithmetic compares integers through their remainders. It lets us simplify large calculations and provides the arithmetic used in cryptography. The Euclidean algorithm computes greatest common divisors without factoring.

All numbers are integers unless stated otherwise. We use the following notation.

Natural numbers	$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$
All integers	$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$
Positive integers	$\mathbb{N}^+ = \mathbb{Z}^+ = \{1, 2, 3, \ldots\}$

Integer division expresses a dividend as a multiple of the divisor plus a remainder. For example, $17 = 3 \times 5 + 2$: the quotient is 5 and the remainder is 2.

Theorem 1.1 (Division algorithm)**.** *For every integer a and positive integer d, there are unique integers q, r satisfying*

$$a = dq + r, \qquad 0 \le r < d.$$

Here a is the dividend, d the divisor, q the quotient, and r the remainder. Write $q = a \operatorname{div} d$ and $r = a \bmod d$. For example,

$$17 = 3 \times 5 + 2, \qquad -17 = 3(-6) + 1.$$

Thus $17 \operatorname{div} 3 = 5$, $17 \bmod 3 = 2$, and $-17 \bmod 3 = 1$. Even a negative dividend has a nonnegative remainder under this convention, which fixes $q = \lfloor a/d \rfloor$.

Definition (Divisibility)**.** For integers a, b with $a \neq 0$, write $a \mid b$ if $b = ak$ for some integer k. We say that a divides b, or that a is a divisor of b.

For $a > 0$, this is equivalent to $b \bmod a = 0$. For example, $3 \mid 15$ because $15 = 3 \times 5$, whereas $3 \nmid 17$.

If $d \mid u$ and $d \mid v$, then $d \mid (su + tv)$ for all integers s, t: writing $u = di$ and $v = dj$ gives $su + tv = d(si + tj)$.

2 Congruence

MUST UNDERSTAND Congruence

Definition (Congruence)**.** For a positive integer m,

$$a \equiv b \pmod{m} \iff a \bmod m = b \bmod m \iff m \mid (a-b).$$

Read $a \equiv b \pmod{m}$ as *a is congruent to b modulo m*. Equivalently, $a = km + b$ for some integer k.

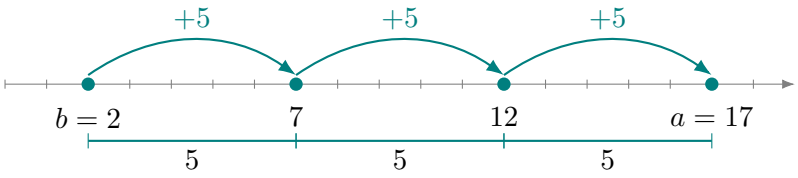
For the latter equivalence, write $a = mq + r$ and $b = mq' + r'$, with $0 \le r, r' < m$. Equal remainders make $a - b$ a multiple of m. Conversely, if $m \mid (a-b)$, then $m \mid (r - r')$. Since $-m < r - r' < m$, this forces $r = r'$.

For example,

$$17 \equiv 2 \pmod{5} \iff 17 \bmod 5 = 2 \bmod 5 \iff 5 \mid (17-2).$$

Both remainders equal 2, and $17 - 2 = 15 = 5 \times 3$.

Starting at b, taking $|k|$ steps of size m reaches a: to the right if $k > 0$, to the left if $k < 0$, and no steps if $k = 0$. For $b = 2$, $a = 17$, and $m = 5$, there are three steps to the right:



$$17 = 2 + 3 \times 5, \qquad 17 - 2 = 5 + 5 + 5 = 15.$$

MUST UNDERSTAND Arithmetic with congruences

Theorem 2.1 (Arithmetic rules)**.** *Let $m > 0$. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then*

$$a + c \equiv b + d, \qquad a - c \equiv b - d, \qquad ac \equiv bd \pmod{m}.$$

Also $a^n \equiv b^n \pmod{m}$ for every positive integer n.

Proof. Write $a = km + b$ and $c = \ell m + d$ for integers k, ℓ. Then $a \pm c = (k \pm \ell)m + (b \pm d)$, proving the addition and subtraction rules. For multiplication, we show that $ac - bd$ is a multiple of m:

$$\begin{aligned} ac - bd &= (km + b)(\ell m + d) - bd \\ &= k\ell m^2 + kmd + b\ell m \\ &= (k\ell m + kd + b\ell) \times m. \end{aligned}$$

Since $k\ell m + kd + b\ell$ is an integer, $ac \equiv bd \pmod{m}$. Repeated multiplication gives the power rule. □

3 Computing with congruences

3.1 Reducing before calculating

Compute $(38 \times 47 + 25 \times 19) \bmod 12$. The four inputs have remainders $2, 11, 1, 7$, so

$$38 \times 47 + 25 \times 19 \equiv 2 \times 11 + 1 \times 7 = 29 \equiv 5 \pmod{12}.$$

The answer is 5. For exact remainders, the final reduction remains necessary:

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m,$$

and similarly for multiplication.

3.2 Computing modular powers

The proposed rule $a^b \equiv a^{\,b \bmod m} \pmod{m}$ is false in general. For example, $6 \equiv 1 \pmod{5}$, but

$$2^6 = 64 \equiv 4 \pmod{5}, \qquad 2^1 \equiv 2 \pmod{5}.$$

Thus $2^6 \not\equiv 2^1 \pmod{5}$. Instead, repeated squaring reduces intermediate products while preserving the intended exponent. In the next lecture we will see that, for $m > 1$, $b \geq 0$, and $\gcd(a, m) = 1$,

$$a^b \equiv a^{\,b \bmod \varphi(m)} \pmod{m},$$

where $\varphi(m)$ is Euler's totient function. The coprimality condition is essential.

MUST UNDERSTAND Two methods for modular powers

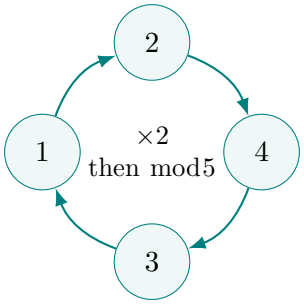
A short repeating cycle can simplify an exponent. Repeated squaring gives a general method that does not require finding such a cycle.

Problem. Compute $2^{13} \bmod 5$.

Solution 1: Build the cycle. Start with $2^1 \bmod 5 = 2$. Multiply the previous remainder by 2 and reduce modulo 5 at each step:

$$\begin{aligned} 2^2 &= 2 \times 2 = 4, \\ 2^3 &\equiv 4 \times 2 = 8 \equiv 3 \pmod{5}, \\ 2^4 &\equiv 3 \times 2 = 6 \equiv 1 \pmod{5}, \\ 2^5 &\equiv 1 \times 2 = 2 \pmod{5}. \end{aligned}$$

The remainder has returned to 2, so the same four-step sequence repeats.



Since $2^4 \equiv 1 \pmod 5$, for every $n \geq 0$,

$$2^{n+4} = 2^n 2^4 \equiv 2^n \pmod 5.$$

Thus four additional factors of 2 leave the remainder unchanged. Since $13 = 3 \times 4 + 1$,

$$2^{13} = (2^4)^3 \times 2 \equiv 1^3 \times 2 = 2 \pmod 5.$$

We reduced the exponent using the verified cycle length 4, not the modulus 5.

More generally, if $a^t \equiv 1 \pmod m$ for some $t > 0$, write a nonnegative exponent $n = qt + r$ with $0 \leq r < t$. Then $a^n = (a^t)^q a^r \equiv a^r \pmod m$. This method is convenient when a short cycle is easy to find. For a general base and modulus, remainders eventually repeat, but there may be an initial nonrepeating part; a return to 1 is not automatic.

Solution 2: Repeated squaring (fast exponentiation). Compute the powers with exponents $1, 2, 4, 8$ by squaring each preceding remainder:

Power	Remainder modulo 5	Computation
2^1	2	starting value
2^2	4	$2^2 = 4$
2^4	1	$4^2 = 16 \equiv 1$
2^8	1	$1^2 = 1$

The binary decomposition $13 = (1101)_2 = 8 + 4 + 1$ tells us which powers to select:

$$2^{13} = 2^8 \times 2^4 \times 2 \equiv 1 \times 1 \times 2 = 2 \pmod 5.$$

In general, express a positive exponent as a sum of distinct powers of two, repeatedly square and reduce, and multiply the selected entries. Unlike searching for a short cycle, this method works directly for any base and positive modulus.

Example: A composite modulus. Compute 3^{14} mod 10 using both a cycle of powers and fast exponentiation. The modulus 10 is composite.

Solution 1. Successive powers have remainders $3, 9, 7, 1, 3, \ldots$. Since $3^4 \equiv 1 \pmod{10}$, this cycle repeats every four powers. Using $14 = 3 \times 4 + 2$,

$$3^{14} = (3^4)^3 \times 3^2 \equiv 1^3 \times 9 = 9 \pmod{10}.$$

Solution 2. Repeated squaring gives

$$3^2 \equiv 9, \qquad 3^4 \equiv 9^2 = 81 \equiv 1, \qquad 3^8 \equiv 1^2 = 1 \pmod{10}.$$

The binary decomposition $14 = (1110)_2 = 8 + 4 + 2$ gives

$$3^{14} = 3^8 \times 3^4 \times 3^2 \equiv 1 \times 1 \times 9 = 9 \pmod{10}.$$

3.3 Cancellation

Multiplication by 2 modulo 6 gives:

x	0	1	2	3	4	5
$2x \bmod 6$	0	2	4	0	2	4

Different inputs can give the same output. In particular, $2 \times 1 \equiv 2 \times 4 \pmod 6$, but $1 \not\equiv 4 \pmod 6$. Multiplication can lose information, so cancellation is not always valid. The next lecture will show that multiplication by a modulo m is reversible precisely when $\gcd(a, m) = 1$.

4 GCDs and the Euclidean algorithm

4.1 Prime factors and common divisors

Definition (Prime and composite integers)**.** An integer $p > 1$ is prime if its only positive divisors are 1 and p. An integer greater than 1 that is not prime is composite.

The first primes are $2, 3, 5, 7, 11, \ldots$. Neither 0 nor 1 is prime or composite.

Factorization means writing a number as a product of factors. For example, $60 = 6 \times 10$. Factoring these further gives

$$60 = 6 \times 10 = (2 \times 3) \times (2 \times 5) = 2^2 \times 3 \times 5.$$

In a *prime factorization*, every factor is prime. A prime cannot be split into smaller positive integer factors: its only positive divisors are 1 and itself.

Theorem 4.1 (Fundamental theorem of arithmetic)**.** *Every integer greater than* 1 *has a unique prime factorization, apart from the order of its factors.*

Existence. Use strong induction on $n \geq 2$. The base case $n = 2$ is prime. For the induction step, if n is prime, it is already a product of one prime. Otherwise, $n = uv$ for some $1 < u, v < n$. By the induction hypothesis, both u and v are products of primes, so n is too. □

The uniqueness proof uses the following lemma.

Lemma 4.2. *If p is prime and neither a nor b is divisible by p, then $ab \not\equiv 0 \pmod p$.*

Proof. Fix a with $p \nmid a$. Suppose a nonzero remainder $b \in \{1, \ldots, p-1\}$ satisfies $ab \equiv 0 \pmod p$, and choose the smallest such b. We have $b \neq 1$, since $p \nmid a$. Divide p by b:

$$p = qb + r, \qquad 0 \leq r < b.$$

If $r = 0$, then b is a divisor of the prime p with $1 < b < p$, which is impossible. Thus $0 < r < b$. But

$$ar = a(p - qb) = ap - q(ab) \equiv 0 \pmod p,$$

contradicting the minimality of b. This proves the lemma for nonzero remainders, hence for all b not divisible by p. Applying it repeatedly shows that a product of factors, none divisible by p, is also not divisible by p. □

Uniqueness of prime factorization. Suppose the same integer has two different prime factorizations. Cancel every common prime factor, including repeated copies, from both sides. If only one side has any factors left, it is greater than 1 while the other side is 1, a contradiction. Otherwise, we have

$$p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s,$$

where the primes on the two sides form disjoint sets. Without loss of generality, let p_1 be the smallest remaining prime and place it on the left. Each q_j is a prime different from p_1, so $p_1 \nmid q_j$. The lemma now gives

$$p_1 p_2 \cdots p_r \equiv 0 \pmod{p_1}, \qquad q_1 q_2 \cdots q_s \not\equiv 0 \pmod{p_1}.$$

Equal integers cannot have different remainders. This contradiction proves that the prime factors, including their exponents, must agree. □

4.2 Greatest common divisors

Definition (Greatest common divisor)**.** For integers a, b not both zero, $\gcd(a, b)$ is the largest positive integer g such that $g \mid a$ and $g \mid b$.

For example,

$$\begin{aligned} 5 &= 5 \times 1, & 10 &= 5 \times 2, & \gcd(5, 10) &= 5, \\ 8 &= 4 \times 2, & 12 &= 4 \times 3, & \gcd(8, 12) &= 4, \\ 60 &= 12 \times 5, & 84 &= 12 \times 7, & \gcd(60, 84) &= 12. \end{aligned}$$

Two integers are *coprime*, or relatively prime, if their GCD is 1. For example, $\gcd(8, 9) = 1$, $\gcd(17, 19) = 1$, and $\gcd(9, 10) = 1$. Coprime integers need not be prime.

The fundamental theorem of arithmetic gives a simple way to find the GCD of positive integers: match their prime factors and take the smaller exponent of each prime. A missing prime has exponent 0. For example,

$$\begin{aligned} 72 &= 2^3 \times 3^2 \times 5^0, \\ 60 &= 2^2 \times 3^1 \times 5^1, \\ \gcd(72, 60) &= 2^{\min(3,2)} \times 3^{\min(2,1)} \times 5^{\min(0,1)} = 12. \end{aligned}$$

A common divisor cannot use more copies of a prime than either integer contains. Taking the smaller exponent for every prime therefore gives the greatest common divisor.

Euclid's method computes a GCD using only division, without requiring either prime factorization.

MUST UNDERSTAND Euclidean reduction

Theorem 4.3 (Euclidean reduction)**.** *If $a \geq 0$, $b > 0$, and $a = bq + r$ with $0 \leq r < b$, then*

$$\gcd(a, b) = \gcd(b, r).$$

Proof. A positive common divisor d of a, b divides $a - qb = r$, so it divides b, r. Conversely, a positive common divisor of b, r divides $qb + r = a$, so it divides a, b. The pairs have exactly the same common divisors and hence the same GCD. □

4.3 The Euclidean algorithm

At each step, replace (a, b) by $(b, a \bmod b)$. This explicitly swaps the old second entry into the first position.

Algorithm 1 Euclidean algorithm

Input: Integers a, b, not both zero.
Output: The greatest common divisor $\gcd(a, b)$.

```
1: procedure EUCLID(a, b)
2:     a ← |a|,    b ← |b|
3:     while b ≠ 0 do
4:         r ← a mod b
5:         a ← b
6:         b ← r
7:     end while
8:     return a
9: end procedure
```

The temporary variable r stores the remainder before either entry is updated.

Example. Find $\gcd(43, 17)$. Repeated division gives

$$\begin{aligned} 43 &= 2 \times 17 + 9, & \gcd(43, 17) &= \gcd(17, 9), \\ 17 &= 1 \times 9 + 8, & &= \gcd(9, 8), \\ 9 &= 1 \times 8 + 1, & &= \gcd(8, 1), \\ 8 &= 8 \times 1 + 0, & &= \gcd(1, 0) = 1. \end{aligned}$$

The same computation as a sequence of pairs is:

$$(43, 17) \longrightarrow (17, 9) \longrightarrow (9, 8) \longrightarrow (8, 1) \longrightarrow (1, 0)$$

Every pair has the same GCD. While the second entry is nonzero, replace (a, b) by $(b, a \bmod b)$. The second entry decreases strictly and stays nonnegative, so the process must terminate. At $(g, 0)$, the GCD is g: every positive divisor of g divides 0, and the greatest is g.

Summary

Congruence expresses equality of remainders. Addition, subtraction, multiplication, and positive integer powers preserve it; cancellation requires extra conditions. Cycles and repeated squaring compute modular powers. Prime factorization describes common divisors, while the Euclidean algorithm finds their GCD without factoring.

5 Exercises

1. Find the quotient and remainder when -29 is divided by 6.
2. Suppose $n \equiv 2 \pmod 5$. Find $(n^2 + n) \bmod 5$.
3. Compute $3^{14} \bmod 10$ using both a cycle and repeated squaring.
4. Compute $\gcd(84, 30)$ using the Euclidean algorithm. Show each pair.
5. Explain why $2x \equiv 2 \pmod 6$ does not imply $x \equiv 1 \pmod 6$. Find all possible remainders of x modulo 6.

Solutions

1. $-29 = 6(-5) + 1$, so the quotient is -5 and the remainder is 1.
2. $n^2 + n \equiv 2^2 + 2 = 6 \equiv 1 \pmod 5$.
3. The cycle is $3, 9, 7, 1$, of length 4. Since $14 \equiv 2 \pmod 4$, the answer is 9. Repeated squaring gives remainders $3, 9, 1, 1$ for exponents $1, 2, 4, 8$; use $14 = 8 + 4 + 2$ to obtain $1 \times 1 \times 9 = 9$.
4. $(84, 30) \to (30, 24) \to (24, 6) \to (6, 0)$, so the GCD is 6.
5. Both $x = 1$ and $x = 4$ give remainder 2 after multiplication by 2. In fact, $6 \mid 2(x - 1)$ is equivalent to $3 \mid (x - 1)$, so the solutions are $x \equiv 1, 4 \pmod 6$.

Reference: Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 8th edition, Sections 4.1–4.3.